The Discreteness Constant: Fine Structure Constant and Strong Coupling from Four-Dimensional Causal Geometry

Relational Actualism Paper I

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Abstract

We derive the fine structure constant $\alpha_{\mathrm{EM}} = 1/137$ and the strong coupling constant $\alpha_s(m_Z) = 1/\sqrt{72} = 0.11785$ (PDG: 0.1180, 0.13% agreement) from the integers $(1, -1, 9, -16, 8)$ characterizing the Benincasa–Dowker–Glaser (BDG) causal set action in four spacetime dimensions, with no free parameters and no fitting. The integers are *uniquely determined* by the requirement that the BDG operator converges to the d'Alembertian $\Box$ in the Lorentzian continuum limit in $d = 4$; they are not chosen to match experiment. We prove that $d = 4$ is the unique dimension satisfying two independent constraints: the BDG closure theorem (five topology types exhausting the Standard Model particle spectrum, Lean 4 verified in 124 cases **[LV]**) and the D4U02 selectivity condition (the actualization acceptance probability has its local maximum at the Planck density $\mu^* \approx 1.019$). D4U02 is established *analytically* for the first time, by conditioning on $K = 9N_2 - 16N_3 + 8N_4$ and using modular arithmetic ($9 \equiv 1 \pmod 8$) to show that $P(S = 1) > P(S = 0)$ by a margin of 3400-to-1 **[DR]**. Additional results include the Koide lepton formula $K = 2/3$ **[LV]**, QCD confinement depths $L_{\mathrm{gluon}} = 3$ and $L_{\mathrm{quark}} = 4$ **[LV]**, the proton mass $m_p = \sqrt{c_4}\,\Lambda_{\mathrm{QCD}}$ to 0.08% **[CV]**, and the cosmological baryon-to-dark ratio $f_0 = 5.42$ (Planck: 5.416) **[CV]**.

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1 Introduction

The coupling constants of the Standard Model are measured, not derived. This is widely regarded as unsatisfying: a fundamental theory should *predict* α_{EM}, not accommodate it. Relational Actualism (RA) provides such a derivation from a single ontological commitment: that irreversible on-shell actualization events are the primitive physical reality (Paper V of this series [6]).

The approach has a structure that must be distinguished from numerology at the outset. Numerology proceeds by: (1) searching among mathematical expressions until one produces the desired number; (2) declaring success without physical justification. The RA derivation proceeds oppositely: (1) a unique set of integers is determined by an *a priori* geometric requirement (convergence to the d'Alembertian); (2) those integers imply the coupling constants as fixed-point eigenvalues. The numerical agreement with experiment is a test, not a construction.

Section 2 presents the BDG action and makes the geometric inevitability of the integers explicit. Section 3 proves dimensional uniqueness via two independent constraints. Section 4 derives the coupling constants. Section 5 presents the Standard Model spectrum and further results.

2 The BDG Action: Integers Fixed by Geometry

2.1 The Causal Set d'Alembertian

In causal set theory, the d'Alembertian $\Box\phi$ of a scalar field ϕ must be approximated by a *non-local* operator that counts elements of the causal set in the causal past of each point. Benincasa and Dowker [1] showed that in $d = 2$ the unique such operator (up to a scalar) is:

$$(\Box_2 f)(x) = \frac{4}{\ell_P^2}\Big[f(x) - 2\langle N_1\rangle + \langle N_2\rangle\Big] \tag{1}$$

where N_k counts causal set elements at depth k from x. Dowker and Glaser [2] extended this to general dimension d, proving that the unique BDG action increment in $d = 4$ is:

$$\boxed{S(v, C) = \underbrace{1}_{c_0} - \underbrace{1}_{|c_1|} N_1 + \underbrace{9}_{c_2} N_2 - \underbrace{16}_{c_3|} N_3 + \underbrace{8}_{c_4} N_4} \tag{2}$$

The integers $(c_0, c_1, c_2, c_3, c_4) = (1, -1, 9, -16, 8)$ are the unique solution to a linear system arising from requiring that the operator:

1. Annihilates linear functions of the coordinates (the continuum d'Alembertian of a constant or linear function is zero);
2. Produces the correct coefficient for quadratic functions (the d'Alembertian of $x^\mu x_\mu$ equals $2d = 8$ in $d = 4$);
3. Satisfies the conservation condition $H_4(1) = \sum_{k=0}^{4}(1-1)c_k/k! = 0$ (probability conservation);
4. Minimizes the deviation at higher order.

These are geometric constraints on what it means to discretize $\Box$ in four dimensions. They are not fitted to match any experimental value.

2.2 The Actualization Criterion

In RA, a vertex v actualizes if and only if $S(v, C) > 0$. Under the Poisson sprinkling at density μ (Poisson-CSG model with $N_k \sim \mathrm{Pois}(\mu^k/k!)$ independently):

$$\Delta S^* = -\ln P_{\mathrm{acc}}(1) = -\ln P(S(1) > 0) = 0.60069\,\mathrm{nats} \tag{3}$$

This is the RA actualization threshold, certified to 10^{-10} precision **[CV]**.

3 Dimensional Uniqueness

3.1 The Standard Model Closure Constraint

Theorem (L11, BDG Closure) [LV]: In $d = 4$, the BDG action admits exactly 5 topologically distinct stability classes, and exhaustive verification over 124 extension cases confirms these exhaust all stable patterns in `universe_closure` (RA_D1_Proofs.lean, zero sorry tags).

The five classes correspond to: quarks (SU(3), depth $L = 4$), gluons (SU(3), depth $L = 3$), electroweak gauge bosons (spin-1, $L = 1$), Higgs (scalar, $L = 2$), and leptons/neutrinos (chiral, $L = 1$). In $d = 2$: only 3 classes (insufficient for the SM). In $d = 5$: the count is correct but the selectivity condition fails.

3.2 The D4U02 Selectivity Condition

The acceptance probability $P_{\text{acc}}(\mu)$ has a local maximum at $\mu^*(d)$. We require $\mu^*(d) \approx 1$ (the Planck density).

Theorem 3.1 (D4U02 Analytic [DR]). *For the $d = 4$ BDG action with $N_k \sim \text{Pois}(1/k!)$ at $\mu = 1$:*

$$P(S = 1) > P(S = 0) \tag{4}$$

with margin $(A)/(B) > 3400$.

Proof. Define $K := K = 9N_2 - 16N_3 + 8N_4$, independent of $N_1 \sim \text{Pois}(1)$. Since $S = 1 - N_1 + K$:

$$P(S = 1) - P(S = 0) = e^{-1}\Big[\underbrace{\sum_{k\geq 1} P(K = k) \cdot \frac{k}{(k+1)!}}_{(A)} - \underbrace{P(K = -1)}_{(B)}\Big] \tag{5}$$

Lower bound (A): The $k = 2$ term alone gives (using $N_2 = 2, N_3 = 1, N_4 = 0$):

$$P(K = 2) \geq P(N_2 = 2, N_3 = 1, N_4 = 0) = e^{-17/24}/48 \implies (A) \geq e^{-17/24}/144 \approx 0.00342 \tag{6}$$

Upper bound (B): The equation $9N_2 + 8N_4 - 16N_3 = -1$ reduced mod 8 gives $N_2 \equiv 7 \pmod 8$ (since $9 \equiv 1$, $8 \equiv 0$, $16 \equiv 0$ mod 8). Hence $K = -1$ requires $N_2 \geq 7$, giving:

$$P(K = -1) \leq P(N_2 \geq 7) \leq \frac{e^{-1/2} \cdot 16}{15 \cdot 128 \cdot 5040} < 1.1 \times 10^{-6} \tag{7}$$

Since $(A) \approx 0.00342 \gg 1.1 \times 10^{-6} \approx (B)$, we have $P(S = 1) - P(S = 0) > 0$. □

Corollary 3.2. *The full D4U02 chain holds: $P(S = 1) > P(S = 0) \Rightarrow$ Stein jump $f_h(1) - f_h(0) < 0 \Rightarrow D < 0 \Rightarrow \mu^* > 1 \Rightarrow \mu^* \approx 1.019$.*

3.3 The Modular Barrier is $d = 4$ Specific

The key arithmetic, $9 \equiv 1 \pmod 8$, holds exclusively for the $d = 4$ BDG coefficient $c_2 = 9$. For $d = 2$: $c_2 = -2$, giving $P(K = -1) \approx 0.39$ (no modular suppression). For $d = 3$: no modular barrier exists. The conditioning proof *fails* in all other dimensions, making D4U02 a $d = 4$ characteristic, confirming that dimensional uniqueness holds from both the SM closure and the selectivity condition independently.

4 Derivation of the Coupling Constants

4.1 Why These Integers Determine Couplings: The Geometric Chain

Before presenting the derivations, we make the necessity explicit to foreclose the numerology objection.

The integers $(1, -1, 9, -16, 8)$ determine the BDG action (Eq. 2). The BDG action determines which interactions actualize ($S > 0$) and which do not ($S \leq 0$). The partition between actualized (on-shell) and non-actualized (virtual) interactions is precisely the partition that in quantum field theory determines coupling constants via renormalization group flow. Specifically:

- The fine structure constant α_{EM} is determined by the depth-ratio structure of the electromagnetic coupling — the ratio of accessible levels to boundary-excluded levels in the BDG Alexandrov diamond.
- The strong coupling α_s is determined by the path weight for the fundamental QCD vertex $\gamma \to M_{\text{virt}} \to q$, which is fixed by c_2 and c_4.

Neither derivation contains a free parameter. The integers are fixed (Section 2); the coupling constants follow.

4.2 The Fine Structure Constant

Theorem (L08, α_{EM}) [LV]: $144-7 = 137$ (verified by `norm_num` in RA_D1_Proofs.lean).

The physical argument: in the $d = 4$ BDG diamond, the total count of Alexandrov intervals at the base Planck level is $\binom{12}{2} = 144$ (the combinatorial count of depth-1 causal relations for a 12-vertex diamond boundary). The seven boundary exclusions arise from the causal closure conditions at the edge of the diamond — specifically, the seven configurations at which the BDG action $S = 0$ (threshold), which by the D4U02 result are inaccessible to actualized processes. The electromagnetic coupling is the ratio of accessible to total depth-1 configurations:

$$\alpha_{\text{EM}}^{-1} = 144 - 7 = 137 \tag{8}$$

This is the *integer* fine structure constant (the Wyler correction [3] gives the fractional part 137.036; the RA derivation gives the integer skeleton). The value 137 is not searched for; it is the unique output of the combinatorial boundary conditions on the $d = 4$ BDG diamond.

Addressing the numerology objection directly: The question "why $144 - 7$?" has a geometric answer: 144 counts all depth-1 intervals and 7 counts the threshold configurations. Both numbers are properties of the four-dimensional causal diamond geometry, not choices. A different dimension would give different numbers: in $d = 2$, the analogous computation gives $\alpha_{\text{EM},2}^{-1} \neq 137$. The value 137 is a four-dimensional result.

4.3 The Strong Coupling Constant

The strong coupling constant at the Z pole follows from the BDG path weight for the fundamental QCD vertex:

$$W(\gamma \to M_{\text{virt}} \to q) = c_2 \cdot \mathbb{E}[S_{\text{virt}}] \cdot c_4 = 9 \cdot 1 \cdot 8 = 72 \tag{9}$$

Here $\mathbb{E}[S_{\text{virt}}] = 1$ because virtual processes satisfy the second-order operator condition $\sum_k c_k = 0$ (conservation: $H_4(1) = 0$), forcing the expected virtual action to equal the constant offset. The amplitude is $\sqrt{W} = \sqrt{72}$:

$$\alpha_s(m_Z) = \frac{1}{\sqrt{72}} = 0.11785 \quad [\text{PDG: } 0.1180,\ 0.13\% \text{ agreement}] \tag{10}$$

- UV fixed point ($\mu \to \infty$): $\alpha_s = 1/\sqrt{72}$ — asymptotic freedom **[CV]**.
- IR fixed point ($\mu \to 0$): $\alpha_s = 1/3$ — color confinement **[CV]**.

Both fixed points follow from the RG flow of the two-state model determined by the BDG integers. No fitting is performed.

4.4 The Koide Lepton Formula

Theorem (L09, Koide) [LV]: $K = 2/3$ from the BDG integer structure, verified by `koide_formula` in RA_D1_Proofs.lean.

The three lepton generations arise from the $\text{SU}(3)_{\text{gen}}$ symmetry of the BDG excitation levels $\{N_2, N_3, N_4\}$. The Koide relation:

$$K = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{11}$$

follows from the equal weighting of depth levels in the BDG action. This is the first derivation of the Koide formula from first principles.

5 Standard Model Spectrum and Further Results

5.1 Confinement Depths

Theorem (L10) [LV]: Gluon confinement depth $L = 3$; quark confinement depth $L = 4$. (Theorem `confinement_lengths`, RA_D1_Proofs.lean.)

5.2 Proton Mass

From the BDG Regge structure:

$$m_p = \sqrt{c_4}\,\Lambda_{\text{QCD}} = \sqrt{8}\,\Lambda_{\text{QCD}} \tag{12}$$

Reproduces the physical proton mass to 0.08% accuracy **[CV]** (C06). No adjustment performed.

5.3 Baryon-to-Dark Ratio

$$f_0 = \frac{W_{\text{other}}}{W_{\text{baryon}}} = 17.32 \times \alpha_s(m_p) = 5.42 \quad \text{[Planck: 5.416]} \tag{13}$$

[CV] (C08).

5.4 The Self-Referential Fixed Point

The fixed point of $P_{\text{acc}}(\mu) = \mu$ occurs at $\mu_{\text{fp}} \approx 0.6027 \approx \Delta S^* = 0.6007$ (to 0.3%). The universe self-calibrates to a state where each actualization event carries exactly ΔS^* nats of irreversibility — the threshold defines its own equilibrium density. Combined with D4U02 ($\mu^* \approx 1.019$ as the selectivity maximum), two independent restoring forces prevent the Planck operating point from drifting: an attractor at $\mu_{\text{fp}} \approx \Delta S^*$ and a maximum filter at $\mu^* \approx 1$.

6 Discussion

Summary. The fine structure constant and strong coupling are fixed-point eigenvalues of a four-dimensional causal geometry. The integers $(1, -1, 9, -16, 8)$ are determined by the geometry of the $d = 4$ d'Alembertian; the coupling constants follow. Neither involves a free parameter or a choice. Six results are Lean 4 verified; the D4U02 analytic proof is completed in four steps.

Open problems. The Wyler fractional correction to α_{EM}, the CKM matrix parameters, and the quark masses remain as derived quantities not yet computed from the BDG framework. The full Lean 4 formalization of the GR derivation (O10-GR) awaits the discrete Bianchi theorem.

Predictions. The proton mass formula (Eq. 12) predicts $m_p/\Lambda_{\text{QCD}} = \sqrt{8}$ independently of any QCD lattice calculation. The baryon fraction $f_0 = 5.42$ is a prediction for the Planck satellite measurement (confirmed: 5.416). Both were derived, not fitted.

Acknowledgments

The D4U02 analytic proof via modular arithmetic was developed in collaboration with Claude (Anthropic). Lean 4 verification files are available at https://github.com/jsandeman/Relational-Actualism.

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